

Table 1: Numerical Results for THMRST1, THMRST2, THMRST3, THMRST4, and THMRST5 on problem 1

#Dim	Ip	THMRST1					THMRST2					THMRST3					THMRST4					THMRST5				
		#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	
1,000	x1	2	4	0.14566	0	2	4	0.00815	0	2	4	0.009887	0	2	4	0.007989	0	2	4	0.009074	0	2	4	0.009074	0	
	x2	1	2	0.005495	0	1	2	0.003664	0	1	2	0.002897	0	1	2	0.002034	0	1	2	0.003034	0	1	2	0.003034	0	
	x3	1	2	0.002832	0	1	2	0.002138	0	1	2	0.002188	0	1	2	0.002061	0	1	2	0.002395	0	1	2	0.002395	0	
	x4	1	4	0.04316	0	1	4	0.010073	0	1	4	0.008422	0	1	4	0.007724	0	1	4	0.007193	0	1	4	0.007193	0	
	x5	1	2	0.018097	0	1	2	0.00806	0	1	2	0.009842	0	1	2	0.007254	0	1	2	0.008445	0	1	2	0.008445	0	
5,000	x1	2	4	0.021789	0	2	4	0.009099	0	2	4	0.008738	0	2	4	0.007038	0	2	4	0.006384	0	2	4	0.006384	0	
	x2	1	2	0.001486	0	1	2	0.001489	0	1	2	0.002679	0	1	2	0.001437	0	1	2	0.001451	0	1	2	0.001451	0	
	x3	1	2	0.001467	0	1	2	0.001481	0	1	2	0.002786	0	1	2	0.001869	0	1	2	0.001396	0	1	2	0.001396	0	
	x4	1	4	0.00158	0	1	4	0.001891	0	1	4	0.002169	0	1	4	0.001514	0	1	4	0.001484	0	1	4	0.001484	0	
	x5	1	2	0.001533	0	1	2	0.001471	0	1	2	0.001607	0	1	2	0.001437	0	1	2	0.001369	0	1	2	0.001369	0	
10,000	x1	2	4	0.003819	0	2	4	0.002778	0	2	4	0.002792	0	2	4	0.002737	0	2	4	0.004311	0	2	4	0.004311	0	
	x2	1	2	0.001946	0	1	2	0.00174	0	1	2	0.001682	0	1	2	0.00173	0	1	2	0.002532	0	1	2	0.002532	0	
	x3	1	2	0.001988	0	1	2	0.001747	0	1	2	0.002451	0	1	2	0.001736	0	1	2	0.001965	0	1	2	0.001965	0	
	x4	1	4	0.003111	0	1	4	0.003135	0	1	4	0.002343	0	1	4	0.002029	0	1	4	0.002947	0	1	4	0.002947	0	
	x5	1	2	0.001929	0	1	2	0.002778	0	1	2	0.001847	0	1	2	0.002184	0	1	2	0.002356	0	1	2	0.002356	0	
50,000	x1	2	4	0.011501	0	2	4	0.010632	0	2	4	0.010204	0	2	4	0.01313	0	2	4	0.01041	0	2	4	0.01041	0	
	x2	1	2	0.005181	0	1	2	0.006503	0	1	2	0.005302	0	1	2	0.006172	0	1	2	0.005859	0	1	2	0.005859	0	
	x3	1	2	0.005715	0	1	2	0.006162	0	1	2	0.004884	0	1	2	0.007507	0	1	2	0.007425	0	1	2	0.007425	0	
	x4	1	4	0.005855	0	1	4	0.008375	0	1	4	0.006948	0	1	4	0.00882	0	1	4	0.005845	0	1	4	0.005845	0	
	x5	1	2	0.005694	0	1	2	0.006065	0	1	2	0.005878	0	1	2	0.007592	0	1	2	0.005874	0	1	2	0.005874	0	
100,000	x1	2	4	0.027806	0	2	4	0.018289	0	2	4	0.025591	0	2	4	0.02627	0	2	4	0.023878	0	2	4	0.023878	0	
	x2	1	2	0.008519	0	1	2	0.010295	0	1	2	0.012592	0	1	2	0.014258	0	1	2	0.01282	0	1	2	0.01282	0	
	x3	1	2	0.012489	0	1	2	0.012092	0	1	2	0.01041	0	1	2	0.011215	0	1	2	0.011148	0	1	2	0.011148	0	
	x4	1	4	0.013556	0	1	4	0.012651	0	1	4	0.016748	0	1	4	0.018123	0	1	4	0.021599	0	1	4	0.021599	0	
	x5	1	2	0.012434	0	1	2	0.010325	0	1	2	0.012726	0	1	2	0.031248	0	1	2	0.010234	0	1	2	0.010234	0	

Table 2: Numerical Results for THMRST1, THMRST2, THMRST3, THMRST4, and THMRST5 on Problem 2

#Dim	Ip	THMRST1					THMRST2					THMRST3					THMRST4					THMRST5				
		#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	
1,000	x1	1	5	0.005086	0	1	5	0.002137	0	1	5	0.002559	0	1	5	0.002994	0	1	5	0.003783	0	1	5	0.003783	0	
	x2	1	4	0.000992	0	1	4	0.000762	0	1	4	0.000779	0	1	4	0.000947	0	1	4	0.000775	0	1	4	0.000775	0	
	x3	1	4	0.000819	0	1	4	0.000763	0	1	4	0.000805	0	1	4	0.00113	0	1	4	0.000709	0	1	4	0.000709	0	
	x4	1	4	0.00077	0	1	4	0.000777	0	1	4	0.000767	0	1	4	0.001015	0	1	4	0.000728	0	1	4	0.000728	0	
	x5	1	4	0.002196	0	1	4	0.001799	0	1	4	0.001661	0	1	4	0.002022	0	1	4	0.001669	0	1	4	0.001669	0	
5,000	x1	1	5	0.001692	0	1	5	0.001482	0	1	5	0.001534	0	1	5	0.001632	0	1	5	0.00186	0	1	5	0.00186	0	
	x2	1	4	0.00136	0	1	4	0.001485	0	1	4	0.001227	0	1	4	0.001289	0	1	4	0.001166	0	1	4	0.001166	0	
	x3	1	4	0.0013	0	1	4	0.001345	0	1	4	0.001137	0	1	4	0.001398	0	1	4	0.001284	0	1	4	0.001284	0	
	x4	1	4	0.001407	0	1	4	0.001307	0	1	4	0.001095	0	1	4	0.001332	0	1	4	0.001183	0	1	4	0.001183	0	
	x5	1	4	0.001517	0	1	4	0.001118	0	1	4	0.001412	0	1	4	0.001362	0	1	4	0.001195	0	1	4	0.001195	0	
10,000	x1	1	5	0.001783	0	1	5	0.00219	0	1	5	0.001849	0	1	5	0.002134	0	1	5	0.002666	0	1	5	0.002666	0	
	x2	1	4	0.002081	0	1	4	0.002516	0	1	4	0.001984	0	1	4	0.002695	0	1	4	0.001791	0	1	4	0.001791	0	
	x3	1	4	0.002619	0	1	4	0.001768	0	1	4	0.002003	0	1	4	0.002202	0	1	4	0.00184	0	1	4	0.00184	0	
	x4	1	4	0.012427	0	1	4	0.001668	0	1	4	0.001657	0	1	4	0.001845	0	1	4	0.001758	0	1	4	0.001758	0	
	x5	1	4	0.002262	0	1	4	0.001815	0	1	4	0.002773	0	1	4	0.001887	0	1	4	0.001973	0	1	4	0.001973	0	
50,000	x1	1	5	0.00666	0	1	5	0.005668	0	1	5	0.006506	0	1	5	0.007553	0	1	5	0.005253	0	1	5	0.005253	0	
	x2	1	4	0.007002	0	1	4	0.004814	0	1	4	0.004465	0	1	4	0.007304	0	1	4	0.005775	0	1	4	0.005775	0	
	x3	1	4	0.005454	0	1	4	0.005197	0	1	4	0.004965	0	1	4	0.004837	0	1	4	0.005458	0	1	4	0.005458	0	
	x4	1	4	0.00532	0	1	4	0.005107	0	1	4	0.006672	0	1	4	0.006548	0	1	4	0.00619	0	1	4	0.00619	0	
	x5	1	4	0.007386	0	1	4	0.005225	0	1	4	0.005885	0	1	4	0.006076	0	1	4	0.0052	0	1	4	0.0052	0	
100,000	x1	1	5	0.010562	0	1	5	0.011663	0	1	5	0.013643	0	1	5	0.017211	0	1	5	0.013064	0	1	5	0.013064	0	
	x2	1	4	0.009241	0	1	4	0.00948	0	1	4	0.009411	0	1	4	0.010951	0	1	4	0.01074	0	1	4	0.01074	0	
	x3	1	4	0.013999	0	1	4	0.0111	0	1	4	0.01295	0	1	4	0.010337	0	1	4	0.01116	0	1	4	0.01116	0	
	x4	1	4	0.009757	0	1	4	0.009832	0	1	4	0.010618	0	1	4	0.014494	0	1	4	0.008551	0	1	4	0.008551	0	
	x5	1	4	0.009271	0	1	4	0.010397	0	1	4	0.011698	0	1	4	0.012675	0	1	4	0.01326	0	1	4	0.01326	0	

Table 3: Numerical Results for THMRST1, THMRST2, THMRST3, THMRST4, and THMRST5 on Problem 3

#Dim	Ip	THMRST1					THMRST2					THMRST3					THMRST4					THMRST5				
		#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	
1,000	x1	1	5	0.008774	0	1	5	0.002629	0	1	5	0.002842	0	1	5	0.002918	0	1	5	0.001853	0	1	5	0.001853	0	
	x2	1	4	0.001067	0	1	4	0.000866	0	1	4	0.000932	0	1	4	0.000936	0	1	4	0.000971	0	1	4	0.000971	0	
	x3	1	4	0.001011	0	1	4	0.000839	0	1	4	0.000848	0	1	4	0.000858	0	1	4	0.000918	0	1	4	0.000918	0	
	x4	1	4	0.00098	0	1	4	0.000838	0	1	4	0.000826	0	1	4	0.000889	0	1	4	0.000904	0	1	4	0.000904	0	
	x5	1	4	0.001524	0	1	4	0.001747	0	1	4	0.001357	0	1	4	0.001046	0	1	4	0.001033	0	1	4	0.001033	0	
5,000	x1	1	5	0.002012	0	1	5	0.002033	0	1	5	0.002040	0	1	5	0.002041	0	1	5	0.001810	0	1	5	0.001810	0	
	x2	1	4	0.001705	0	1	4	0.001699	0	1	4	0.002049	0	1	4	0.001643	0	1	4	0.001686	0	1	4	0.001686	0	
	x3	1	4	0.001688	0	1	4	0.001517	0	1	4	0.001790	0	1	4	0.001677	0	1	4	0.001597	0	1	4	0.001597	0	
	x4	1	4	0.001684	0	1	4	0.001655	0	1	4	0.001631	0	1	4	0.001672	0	1	4	0.001652	0	1	4	0.001652	0	
	x5	1	4	0.001616	0	1	4	0.001531	0	1	4	0.001665	0	1	4	0.001587	0	1	4	0.001573	0	1	4	0.001573	0	
10,000	x1	1	5	0.003775	0	1	5	0.003404	0	1	5	0.002714	0	1	5	0.002698	0	1	5	0.002652	0	1	5	0.002652	0	
	x2	1	4	0.002682	0	1	4	0.004514	0	1	4	0.002419	0	1	4	0.002359	0	1	4	0.002629	0	1	4	0.002629	0	
	x3	1	4	0.002914	0	1	4	0.003014	0	1	4	0.003542	0	1	4	0.002384	0	1	4	0.002497	0	1	4	0.002497	0	
	x4	1	4	0.002997	0	1	4	0.002379	0	1	4	0.002805	0	1	4	0.002488	0	1	4	0.002747	0	1	4	0.002747	0	
	x5	1	4	0.002259	0	1	4	0.003164	0	1	4	0.002450	0	1	4	0.002998	0	1	4	0.002380	0	1	4	0.002380	0	
50,000	x1	1	5	0.012544	0	1	5	0.011922	0	1	5	0.014771	0	1	5	0.015111	0	1	5	0.013415	0	1	5	0.013415	0	
	x2	1	4	0.012961	0	1	4	0.011921	0	1	4	0.010671	0	1	4	0.012653	0	1	4	0.012301	0	1	4	0.012301	0	
	x3	1	4	0.011342	0	1	4	0.009992	0	1	4	0.012424	0	1	4	0.010450	0	1	4	0.012985	0	1	4	0.012985	0	
	x4	1	4	0.012718	0	1	4	0.011943	0	1	4	0.011553	0	1	4	0.011600	0	1	4	0.011803	0	1	4	0.011803	0	
	x5	1	4	0.011580	0	1	4	0.009676	0	1	4	0.010029	0	1	4	0.013617	0	1	4	0.016055	0	1	4	0.016055	0	
100,000	x1	1	5	0.030937	0	1	5	0.026972	0	1	5	0.035189	0	1	5	0.038267	0	1	5	0.027779	0	1	5	0.027779	0	
	x2	1	4	0.027752	0	1	4	0.022732	0	1	4	0.023591	0	1	4	0.026466	0	1	4	0.017795	0	1	4	0.017795	0	
	x3	1	4	0.022811	0	1	4	0.022077	0	1	4	0.022784	0	1	4	0.030382	0	1	4	0.017040	0	1	4	0.017040	0	
	x4	1	4	0.022070	0	1	4	0.020343	0	1	4	0.021639	0	1	4	0.021357	0	1	4	0.024077	0	1	4	0.024077	0	
	x5	1	4	0.024667	0	1	4	0.022437	0	1	4	0.025222	0	1	4	0.025324	0	1	4	0.024191	0	1	4	0.024191	0	

Table 4: Numerical Results for THMRST1, THMRST2, THMRST3, THMRST4, and THMRST5 on Problem 4

#Dim	Ip	THMRST1					THMRST2					THMRST3					THMRST4					THMRST5				
		#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	
1,000	x1	2	9	0.022559	0	2	9	0.002667	0	2	9	0.002500	0	2	8	0.003344	0	2	9	0.003292	0	2	9	0.003292	0	
	x2	2	9	0.001333	0	2	9	0.001211	0	2	9	0.001050	0	2	8	0.001024	0	2	9	0.001241	0	2	9	0.001241	0	
	x3	2	9	0.001103	0	2	9	0.001120	0	2	9	0.001075	0	2	9	0.002050	0	2	9	0.002253	0	2	9	0.002253	0	
	x4	2	9	0.001049	0	2	9	0.001111	0	2	9	0.001124	0	2	8	0.001048	0	2	9	0.001082	0	2	9	0.001082	0	
	x5	2	9	0.002124	0	2	9	0.001274	0	2	9	0.001427	0	2	8	0.001192	0	2	9	0.001908	0	2	9	0.001908	0	
5,000	x1	2	9	0.002472	0	2	9	0.002361	0	2	9	0.002372	0	2	9	0.002414	0	2	9	0.002152	0	2	9	0.002152	0	
	x2	2	9	0.002328	0	2	9	0.002340	0	2	9	0.002338	0	2	8	0.002307	0	2	9	0.002200	0	2	9	0.002200	0	
	x3	2	9	0.002284	0	2	9	0.002238	0	2	9	0.002260	0	2	8	0.002046	0	2	9	0.002211	0	2	9	0.002211	0	
	x4	2	9	0.002501	0	2	9	0.002077	0	2	9	0.002346	0	2	8	0.002344	0	2	9	0.002813	0	2	9	0.002813	0	
	x5	2	9	0.002002	0	2	9	0.002361	0	2	9	0.002277	0	2	8	0.002125	0	2	9	0.002676	0	2	9	0.002676	0	
10,000	x1	2	9	0.003355	0	2	9	0.003227	0	2	9	0.003491	0	2	8	0.003329	0	2	9	0.003158	0	2	9	0.003158	0	
	x2	2	9	0.004836	0	2	9	0.003560	0	2	9	0.003295	0	2	8	0.003543	0	2	9	0.003965	0	2	9	0.003965	0	
	x3	2	9	0.003314	0	2	9	0.004280	0	2	9	0.003345	0	2	8	0.003174	0	2	9	0.003245	0	2	9	0.003245	0	
	x4	2	9	0.003910	0	2	9	0.003642	0	2	9	0.004873	0	2	8	0.003574	0	2	9	0.003333	0	2	9	0.003333	0	
	x5	2	9	0.004071	0	2	9	0.003366	0	2	9	0.003846	0	2	8	0.003910	0	2	9	0.003731	0	2	9	0.003731	0	
50,000	x1	2	9	0.017027	0	2	9	0.013902	0	2	9	0.014265	0	2	8	0.016510	0	2	9	0.011759	0	2	9	0.011759	0	
	x2	2	9	0.016041	0	2	9	0.014702	0	2	9	0.014650	0	2	9	0.014715	0	2	9	0.011754	0	2	9	0.011754	0	
	x3	2	9	0.015399	0	2	9	0.016718	0	2	9	0.014626	0	2	8	0.013556	0	2	9	0.012017	0	2	9	0.012017	0	
	x4	2	9	0.014648	0	2	9	0.015537	0	2	9	0.017513	0	2	8	0.015623	0	2	9	0.015884	0	2	9	0.015884	0	
	x5	2	9	0.011275	0	2	9	0.014634	0	2	9	0.013535	0	2	9	0.015094	0	2	9	0.015399	0	2	9	0.015399	0	
100,000	x1	2	9	0.032522	0	2	9	0.025664	0	2	9	0.032315	0	2	8	0.027024	0	2	9	0.031281	0	2	9	0.031281	0	
	x2	2	9	0.023090	0	2	9	0.031118	0	2	9	0.027082	0	2	8	0.027809	0	2	9	0.027766	0	2	9	0.027766	0	
	x3	2	9	0.031551	0	2	9	0.031504	0	2	9	0.029101	0	2	8	0.024804	0	2	9	0.027853	0	2	9	0.027853	0	
	x4	2	9	0.027276	0	2	9	0.031307	0	2	9	0.035169	0	2	9	0.031033	0	2	9	0.024857	0	2	9	0.024857	0	
	x5	2	9	0.029024	0	2	9	0.023567	0	2	9	0.028117	0	2	8	0.024480	0	2	9	0.024668	0	2	9	0.024668	0	

Table 5: Numerical Results for THMRST1, THMRST2, THMRST3, THMRST4, and THMRST5 on Problem 5

#Dim	Ip	THMRST1					THMRST2					THMRST3					THMRST4					THMRST5				
		#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	#Itr	#Fev	#Tm	#Nrm	
1,000	x1	1	5	0.012517	0	1	5	0.002280	0	1	5	0.003013	0	1	5	0.001956	0	1	5	0.002710	0	1	5	0.002710	0	
	x2	1	4	0.001444	0	1	4	0.000815	0	1	4	0.000889	0	1	4	0.001005	0	1	4	0.000792	0	1	4	0.000792	0	
	x3	1	4	0.000782	0	1	4	0.000792	0	1	4	0.000830	0	1	4	0.000837	0	1	4	0.000797	0	1	4	0.000797	0	
	x4	1	4	0.000999	0	1	4	0.000884	0	1	4	0.000843	0	1	4	0.000891	0	1	4	0.000771	0	1	4	0.000771	0	
	x5	1	4	0.001118	0	1	4	0.001787	0	1	4	0.000995	0	1	4	0.001546	0	1	4	0.001059	0	1	4	0.001059	0	
5,000	x1	1	5	0.001455	0	1	5	0.001287	0	1	5	0.001400	0	1	5	0.001513	0	1	5	0.001419	0	1	5	0.001419	0	
	x2	1	4	0.001333	0	1	4	0.001330	0	1	4	0.001300	0	1	4	0.001289	0	1	4	0.001199	0	1	4	0.001199	0	
	x3	1	4	0.001230	0	1	4	0.001201	0	1	4	0.001166	0	1	4	0.001407	0	1	4	0.001308	0	1	4	0.001308	0	
	x4	1	4	0.001348	0	1	4	0.001138	0	1	4	0.001365	0	1	4	0.001355	0	1	4	0.001168	0	1	4	0.001168	0	
	x5	1	4	0.001278	0	1	4	0.001305	0	1	4	0.001120	0	1	4	0.001115	0	1	4	0.001287	0	1	4	0.001287	0	
10,000	x1	1	5	0.001881	0	1	5	0.001993	0	1	5	0.001953	0	1	5	0.001939	0	1	5	0.013407	0	1	5	0.013407	0	
	x2	1	4	0.002271	0	1	4	0.001906	0	1	4	0.001823	0	1	4	0.001897	0	1	4	0.001667	0	1	4	0.001667	0	
	x3	1	4	0.001906	0	1	4	0.001773	0	1	4	0.001593	0	1	4	0.002605	0	1	4	0.001729	0	1	4	0.001729	0	
	x4	1	4	0.001985	0	1	4	0.001730	0	1	4	0.002035	0	1	4	0.001995	0	1	4	0.001712	0	1	4	0.001712	0	
	x5	1	4	0.001748	0	1	4	0.001939	0	1	4	0.002001	0	1	4	0.002053	0	1	4	0.001631	0	1	4	0.001631	0	
50,000	x1	1	5	0.006181	0	1	5	0.006380	0	1	5	0.006916	0	1	5	0.005688	0	1	5	0.006637	0	1	5	0.006637	0	
	x2	1	4	0.004911	0	1	4	0.007574	0	1	4	0.006058	0	1	4	0.006006	0	1	4	0.004785	0	1	4	0.004785	0	
	x3	1	4	0.007875	0	1	4	0.007828	0	1	4	0.005685	0	1	4	0.004804	0	1	4	0.004587	0	1	4	0.004587	0	
	x4	1	4	0.006975	0	1	4	0.005578	0	1	4	0.005882	0	1	4	0.006015	0	1	4	0.006881	0	1	4	0.006881	0	
	x5	1	4	0.004813	0	1	4	0.005057	0	1	4	0.007805	0	1	4	0.005641	0	1	4	0.004583	0	1	4	0.004583	0	
100,000	x1	1	5	0.011058	0	1	5	0.011283	0	1	5	0.013230	0	1	5	0.013834	0	1	5	0.011468	0	1	5	0.011468	0	
	x2	1	4	0.008184	0	1	4	0.013514	0	1	4	0.010193	0	1	4	0.011631	0	1	4	0.012801	0	1	4	0.012801	0	
	x3	1	4	0.009740	0	1	4	0.012789	0	1	4	0.009331	0	1	4	0.011716	0	1	4	0.011072	0	1	4	0.011072	0	
	x4	1	4	0.013236	0	1	4	0.012771	0	1	4	0.009119	0	1	4	0.009196	0	1	4	0.010695	0	1	4	0.010695	0	
	x5	1	4	0.010410	0	1	4	0.013099	0	1	4	0.011018	0	1	4	0.015026	0	1	4	0.009549	0	1	4	0.009549	0	